F1 Strategy Visualisation

Group Data Visualisation Project Data Audit

We audited FastF1 data for the 2026 Austrian Grand Prix at the Red Bull Ring. The comparison contains 426 driver laps with complete core timing and tyre fields. We retain every selected record and mark 361 laps as eligible for initial normal-racing pace analysis. This audit records preparation decisions, not final performance findings.

Source and provenance

Data were accessed using FastF1 3.8.3 and pandas 2.3.3. The raw download contains 1,339 field laps, results, weather, track status and race-control messages. Telemetry was not downloaded. Raw files remain locally; prepared files and the script are shared in the repository. A SHA-256 hash identifies the raw lap snapshot.

Selected drivers and coverage

Team	Driver	Recorded laps	Pace eligible
Ferrari	HAM	71	61
Ferrari	LEC	71	58
Mercedes	ANT	71	61
Mercedes	RUS	71	60
Red Bull Racing	HAD	71	61
Red Bull Racing	VER	71	60

Missing values and duplicates

No selected lap lacks lap time, compound, tyre age, position, completion time or stint. There are no duplicate driver-lap keys. Sector 1 is missing on six first laps; sectors 2 and 3 are complete. Empty pit timestamps normally mean no pit event on that lap. They are not filled or counted as general missing-data defects.

Structure and variable types

Each main row represents one recorded driver lap. Track-status, weather and race-control tables contain separate timestamped events. The initial analysis does not use spatial variables. Q denotes quantitative, O ordinal, N nominal and T temporal data.

Attributes	Type	Interpretation
Driver, DriverNumber, Team	N	Identifiers; car numbers are categories
LapNumber, Position	O	Ordered lap sequence and race position
Stint	N	Tyre-stint identifier within a driver's race
Compound, FreshTyre	N	Tyre category and fresh-set flag
TyreLife	Q	Age in laps, not measured wear
LapTime and sector times	Q	Durations converted to seconds
Time, LapStartTime, pit times	T	Session-relative timestamps
TrackStatus and quality flags	N	Several status codes can occur on one lap
Derived lap-boundary interval	Q	Same-lap timestamp difference, not a live gap

Unusual values and validation

Lap times range from 70.374 to 113.381 seconds. Slow laps are retained for inspection. Checks confirm positive durations, increasing lap order, unique keys, source row retention and sector totals within 3 milliseconds on eligible laps. All 14 recorded pit entries pair with a later exit for the same driver.

Review of the 55 accuracy flags

FastF1's accuracy logic considers pit activity, track status, sector availability and timing consistency. A false flag does not alone mean a bad record. These mutually exclusive categories account for all 55 flags; pit activity takes priority where categories overlap.

Category	Laps	Treatment
Pit-entry or pit-exit laps	28	Retain for strategy; exclude from normal pace
First laps	6	Retain; exclude standing-start laps from normal pace
Other status-affected laps	21	Retain and annotate; exclude from normal pace
Outside these categories	0	No residual flagged records in this snapshot

Initial pace eligibility rule

An eligible lap must be accurate, green-only and positive in duration, and neither a first lap nor a pit-entry or pit-exit lap. Deleted and generated laps are excluded. This retains 361 laps and excludes 65. The source contains six deleted laps and no generated laps. Status and validity checks explain the difference between 55 accuracy flags and 65 pace exclusions.

Cleaning and transformations

No values were imputed, duplicates deleted or slow laps trimmed. We select the three teams, order records by driver and lap, convert timings to seconds, and add flags and exclusion reasons. There are 25 VSC-affected and 11 yellow-affected selected laps; these counts overlap. No selected lap contains a full Safety Car flag. Lap-level flags do not specify exact event start or end times.

Prepared outputs and aggregation

The selected-lap file retains all 426 records. Separate pace and flagged files support different tasks. Stint summaries record driver-level boundaries, compounds and eligible counts; teammates are not averaged together. Pit visits pair entry and exit times. Their 21.042 to 22.742 second intervals measure elapsed pit-lane time, not stationary service time or the causal cost of stopping.

Prepared dataset preview

Driver	Lap	Compound	Tyre age	Lap seconds
ANT	3	MEDIUM	3	72.568
HAD	2	MEDIUM	2	73.413
HAM	2	MEDIUM	2	72.028

Assumptions and limitations

The derived gap compares like-for-like lap completions, not live positions. Excluding unaligned driver 77 leaves every selected interval unchanged. Driver 63's 0.530-second warning compares his finish with a later session-status message; it is not a timing correction. Selected finishing intervals match the published classification. Intermediate checks share the retained timing source. See the repository timing investigation for evidence and limits.

Fuel load, traffic, weather and track evolution remain confounders. The pace rule is a starting point; stint phase and compound need further checks. Offline reprocessing and 22 tests passed. Human review by Roy and exploratory analysis remain pending.

Reproduction and sources

Run scripts/prepare_race.py, then use --from-snapshot for offline reprocessing. Run python -m unittest discover -s tests -v after downloading. The repository README explains the outputs.

Repository: https://github.com/haydenteh5526/F1-Strategy-Visualisation
FastF1: https://docs.fastf1.dev/core.html
Official result: https://www.formula1.com/en/results/2026/races/1288/austria/race-result